GROWTH HORMONE & ANABOLIC SIGNALING

Sermorelin — the full review

GROWTH HORMONE-RELEASING HORMONE ANALOGUE

What it is

Sermorelin is a lab-made copy of the first 29 building blocks of a natural brain hormone, growth hormone-releasing hormone (GHRH), which signals the pituitary gland to release its own growth hormone in short bursts rather than delivering growth hormone directly.

A version was once sold in the United States under the brand name Geref — first as a single-dose diagnostic test of pituitary function, later as a nightly injection for children with diagnosed growth hormone deficiency. Both approvals were withdrawn in 2009 when the manufacturer discontinued the product for business reasons.

What is available today is a compounded preparation made from bulk sermorelin powder; it has not itself been reviewed by the FDA for safety, effectiveness, or quality, and it only works if the pituitary gland still has growth hormone available to release.

What people use it for

USED FOR	EVIDENCE	WHAT THAT MEANS
Growth failure in children with diagnosed GH deficiency	**FDA-approved**	The one use that ever carried FDA approval — since withdrawn in 2009 for business reasons, not safety.
Adult GH optimization, body composition	**Limited human data**	A six-month study found modest fat and lean-mass changes, but no improvement in strength or fitness.
Recovery, sleep, libido, athletic performance	**Theoretical**	No human trial has tested sermorelin for any of these uses.
Cognitive function	**Limited human data**	Rests on a single six-month, 89-person trial from 2006 using surrogate cognitive tests; never replicated.
Combined with ipamorelin, CJC-1295, or other growth hormone secretagogues	**Theoretical**	Regulatory review found few human studies examining these combinations; the 'synergy' claim comes from practitioner reports, not trial data.

How it works

A hormone from the hypothalamus, GHRH, is the natural “go” signal that prompts the pituitary gland to release growth hormone. Sermorelin is a shortened synthetic copy of that signal; injected under the skin, typically at bedtime, it prompts a pulse of growth hormone release, which then travels to the liver and prompts it to produce IGF-1.

Because sermorelin acts upstream of growth hormone itself, it only works if the pituitary still has growth hormone available to release — if the pituitary is damaged, missing, or unresponsive, sermorelin has nothing to stimulate. It also clears from the body within minutes, producing a short, pulse-like effect rather than a steady elevation.

The idea that this makes it inherently safer than growth hormone itself — because the body's own feedback loops limit how much can be released — is a plausible mechanism, not a finding confirmed by a study that directly compared the two in adults.

What the research actually looked at

The pediatric approval rested on a 12-month, open-label trial without a placebo or comparison group, in previously untreated children with diagnosed growth hormone deficiency; a later review of the pediatric literature found sermorelin's growth effect less pronounced than that of growth hormone itself.

In adults — the population being sold sermorelin today — the published record is small: roughly 130 adults total across a handful of studies, the largest around 90 people, none longer than six months. One six-month study found a measurable rise in IGF-1 and modest shifts in body composition, but no improvement in strength, aerobic fitness, or physical function.

No adult trial has tested sermorelin for fat loss, athletic performance, injury recovery, sleep, libido, or longevity as a primary outcome. The larger and more heavily studied growth hormone literature, which sermorelin's own mechanism depends on, has similarly found no strength or functional benefit from raising growth hormone in healthy older adults.

Common misunderstandings

- **“Sermorelin is FDA-approved.”** Two sermorelin products were approved, in 1990 and 1997, and both approvals were withdrawn in 2009. Nothing containing sermorelin is currently an approved, marketed drug.
- **“It was withdrawn for business reasons, so regulators consider it safe.”** That determination only established that the withdrawal wasn't a safety action — it says nothing about the safety of a compounded, unapproved product at adult doses used for years.
- **“It's a safer, more natural alternative to growth hormone because the body's own feedback loop prevents overdose.”** This is a plausible-sounding mechanism, not a finding from a study — no adult trial has directly compared sermorelin to growth hormone on safety outcomes.
- **“It builds muscle, increases strength, and improves recovery.”** The one relevant six-month adult study found no improvement in strength or aerobic fitness despite a real gain in lean mass, and the much larger growth hormone literature found the same pattern.

Who it is not for

- Anyone whose growth hormone deficiency originates at the pituitary itself, or from a known or suspected pituitary lesion — there is nothing for sermorelin to stimulate, and imaging should come first.
- Anyone with prediabetes, diabetes, or insulin resistance.
- Anyone with active cancer or a recent cancer history.
- Pregnant or breastfeeding patients, or anyone trying to conceive.
- Children and adolescents outside a diagnosed, specialist-managed growth hormone deficiency.
- Anyone unwilling or unable to attend monitoring visits and lab draws.

Competitive athletes: Prohibited at all times, in and out of competition, under the WADA category covering growth hormone-releasing factors. A therapeutic use exemption is considered highly unlikely to be approved, since there is no FDA-approved product or indication to support one for adult use.

Who makes this, and under what rules

Everything dispensed here is prepared by a licensed 503A compounding pharmacy — not bought from a supplier and repackaged, and not sourced from an online seller.

That designation carries specific, enforceable obligations:

- **A prescription in your name.** Compounded from a prescription for one identified patient, after an evaluation.
- **Licensed and inspected.** A state licence, subject to inspection by the state board of pharmacy.
- **Ingredients that can be traced.** A certificate of analysis from an FDA-registered facility, back to a named source and lot.
- **Sterile preparation to a published standard.** USP standards.
- **A paper trail.** A lot number and beyond-use date on every preparation.

A compounded preparation is not an FDA-approved medicine — but it is the difference between a licensed pharmacy working to a standard and an unregulated website.

Where this stands

Once approved, now withdrawn — for business reasons, not safety. Two sermorelin products held FDA approval, in 1990 and 1997. Both approvals were withdrawn in 2009 after the manufacturer stopped making the ingredient; regulators later noted the withdrawal wasn't a safety or effectiveness finding, but that determination doesn't extend to today's compounded, unapproved product at adult doses.

Fewer than about 130 adults have ever been studied, and no trial ran longer than six months. The one relevant body-composition study found a rise in IGF-1 and a lean-mass shift, but no improvement in strength, aerobic fitness, or physical function — a pattern that matches the much larger growth hormone literature.

It only works if the pituitary still has growth hormone to release — a mechanism-level limit, not a study finding.

Clinical detail

Held two FDA approvals — Geref Diagnostic (1990) for a single-dose pituitary function test, and Geref (1997) for growth failure in children with diagnosed growth hormone deficiency; both were formally withdrawn in June 2009 after the manufacturer discontinued the product.

Acts at the GHRH receptor on pituitary somatotrophs, prompting pulsatile release of the body's own growth hormone rather than delivering growth hormone directly.

Does not appear on FDA's 503A bulk-substance category lists, unlike ipamorelin, which is flagged as a Category 2 safety-risk substance — a comparatively more favorable regulatory position within this catalog.

The pediatric approval trial was open-label and uncontrolled; a later review found sermorelin's growth effect less pronounced than growth hormone itself.

The largest published adult trial enrolled roughly 90 people over six months; no adult sermorelin trial has run longer.

The information on this page is general education. It is not medical advice and it is not a recommendation for any treatment.

Sermorelin is not an FDA-approved medicine for the uses described here. A compounded preparation is not an FDA-approved product and has not been reviewed by the FDA for safety, effectiveness or quality.

Descriptions of what this is used for reflect how it is commonly used and discussed. They are not claims that it is effective for any purpose.

Any prescription follows an individual evaluation by a licensed provider and is issued only where that provider judges it clinically appropriate. A prescription is never guaranteed.